Department of Mechanical and Production Engineering

Aarhus University

Formula Sheet

Author: Noah Rahbek Bigum Hansen

Student ID: 202405538

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Contents

1 Constant Definitions

1.1 Coulomb Constant

$$k = \frac{1}{4\pi\epsilon_0} = 8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}.$$

Symbols

k	Coulomb constant	$8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}$
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N\,m^2}$

2 Electrostatics

2.1 Coulomb's Law

$$\boldsymbol{F} = k\frac{q_1 q_2}{r^2}\,\hat{r}.$$

Assumptions

- Point charges

Symbols

$\boldsymbol{F}$	Electrostatic force	N
k	Coulomb constant	$8{,}99 \cdot 10^9\,\mathrm{N\,m^2/C^2}$
q_1, q_2	Charges	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.1.1 Principle of Superposition

$$\boldsymbol{F}_{1,\mathrm{net}} = \sum_{i=1}^{n} \boldsymbol{F}_{1i}.$$

Symbols

$\boldsymbol{F}_{1,\mathrm{net}}$	Net force	N
$\boldsymbol{F}_{1i}$	Force contribution on particle 1 from particle i	N

2.2 Quantization of Charge

$$q = \pm n \cdot e.$$

Symbols

q	Charge	C
n	Integer number ($n \in \mathbb{Z}$)	—
e	Elementary charge	$1{,}6 \cdot 10^{-19}\,\mathrm{C}$

2.3 Electric Field at a point

$$\boldsymbol{E}(x,y,z) = \frac{\boldsymbol{F}(x,y,z)}{q_0}.$$

Symbols

$\boldsymbol{E}(x,y,z)$	Electric field at point (x,y,z)	$\frac{\text{N}}{\text{C}}$
$\boldsymbol{F}(x,y,z)$	Force on test particle at point (x,y,z)	N
q_0	Test charge	C

2.3.1 Electric Field Produced by Point Charge

$$\boldsymbol{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$$
$$E = \frac{1}{4\pi\epsilon_0}\frac{|q|}{r^2}.$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8{,}85\cdot10^{-12}\,\text{C}^2/\text{N}\,\text{m}^2$
q	Charge	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.3.2 Electric Field due to a Dipole along dipole axis

$$E(z) = E_{(+)}(z) + E_{(-)}(z) = \frac{q}{4\pi\epsilon_0 z^2}\left(\frac{1}{1-\frac{d^2}{2z}} - \frac{1}{1+\frac{d^2}{2z}}\right) = \frac{q}{2\pi\epsilon_0 z^3}\left(\frac{d}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2}\right).$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
$E_{(+)}, E_{(-)}$	Contribution to E from the positive and negative part of the dipole	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8{,}85\cdot10^{-12}\,\text{C}^2/\text{N}\,\text{m}^2$
z	Distance along axis	m
d	Separation distance	m
q	Charge	C

2.3.3 Electric Field due to a Dipole along Dipole axis at Long Distance

$$E = \frac{1}{2\pi\epsilon_0}\frac{qd}{z^3} = \frac{1}{2\pi\epsilon_0}\frac{p}{z^3}.$$

Assumptions

- $z \gg d$, i.e. the distance between charges is much smaller then the distance from the charges you are measuring ("normal" case)

Symbols

E	Electric field strength	$\frac{N}{C}$
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,C^2/N\,m^2$
z	Distance along dipole axis	m
d	Separation distance	m
q	Charge	C
p	Electric dipole moment ($p = qd$)	C m

2.3.4 Electric Field of a Charged Ring at Large Distance

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{z^2}.$$

Assumptions

- $z \gg R$, i.e. distance from the ring is much grater than the radius of the ring

Symbols

E	Electric field strength	$\frac{N}{C}$
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,C^2/N\,m^2$
q	Size of charge in the ring	C
z	Distance perpendicular to the radius of the ring	m

2.3.5 Electric Field of a Charged Disk

$$E = \frac{\sigma}{2\epsilon_0}\left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right).$$

Assumptions

- Infinitesimally thin disk

Symbols

E	Electric field strength	$\frac{N}{C}$
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,C^2/N\,m^2$
σ	Surface charge density	C/m^2
z	Distance perpendicular to the disk	m
R	radius of disk	m

2.3.6 Electric field from an infinite sheet

$$E = \frac{\sigma}{2\epsilon_0}.$$

Assumptions

- Infinitesimally thin infinite sheet

Symbols

E	Electric field strength	$\frac{N}{C}$
σ	Surface charge density	C/m^2
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,C^2/N\,m^2$

2.3.7 Electric field of a Conducting Sphere

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = k\frac{q}{r^2}.$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of a conducting sphere.
- Spherical symmetry
- No charge enclosed in conductor

Symbols

E	Electric field strength	$\frac{\mathrm{N}}{\mathrm{C}}$
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N\,m}^2$
q	Charge	C
r	Distance	m
k	Coulomb constant	$8{,}99 \cdot 10^{9}\,\mathrm{N\,m}^2/\mathrm{C}^2$

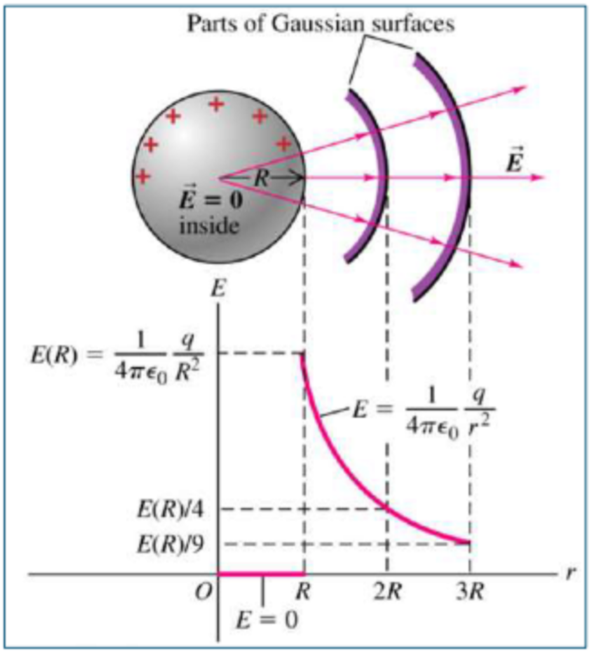

Figure 1: E-field propagation from a conducting sphere.

2.4 Electric Field of an Insulating Sphere

$$\text{Outside}: \quad E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$$
$$\text{Inside}: \quad E = \frac{q}{4\pi\epsilon_0 R^3}r.$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of an insulating sphere.
- Spherical symmetry

Symbols

E	Electric field strength	$\frac{\mathrm{N}}{\mathrm{C}}$
r	Distance	m
R	Radius	m
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N\,m}^2$
q	Charge	C

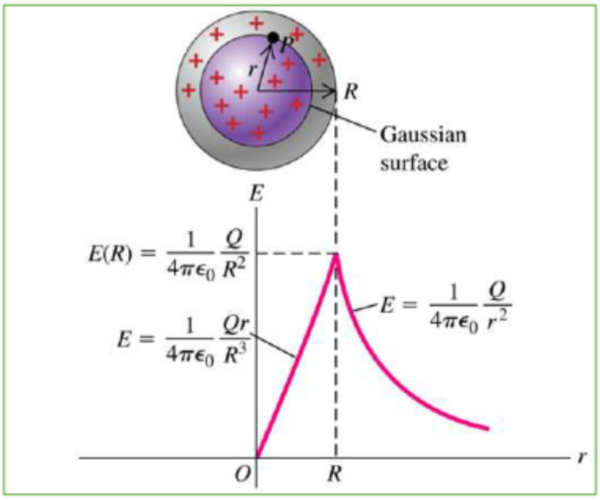

Figure 2: E-field propagation through an insulating sphere.

2.5 Gauss' Law

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0}.$$

Assumptions

- Vacuum (air)

Symbols

Φ	Electric flux	V m
q_{enc}	Enclosed charge	C
ϵ_0	Permittivity constant	$8{,}85 \cdot 10^{-12}\,\mathrm{C}^2/\mathrm{N\,m}^2$

2.5.1 Net electric flux through a surface caused by an electric field

$$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A}.$$

Symbols

Φ	Electric flux	V m
$\boldsymbol{E}$	Electric field vector	$\frac{\mathrm{N}}{\mathrm{C}}$
$\boldsymbol{A}$	Area vector	m^2

2.6 Electric Potential Energy

$$U_E = q_0 E \Delta d.$$

Symbols

U_E	Electric potential energy	J
q_0	Test charge	C
E	Electric field strength	$\frac{\mathrm{N}}{\mathrm{C}}$
d	Separation distance	m

2.6.1 Electrostatic Potential

$$V_E = \frac{U_E}{q_0}.$$

Symbols

V_E	Electrostatic potential	V
U_E	Electric potential energy	J
q_0	Test charge	C

2.6.2 Electrostatic Potential of a Plate Capacitor

$$V_E = Ed.$$

Symbols

V_E	Electrostatic potential	V
E	Electric field strength	$\frac{\mathrm{N}}{\mathrm{C}}$
d	Separation distance	m

3 Fundamentals of Electric Circuits

3.1 Current, Voltage, and Resistance

3.1.1 Electric Current

$$I = \frac{\mathrm{d}q}{\mathrm{d}t}.$$

Symbols

I	Current	A
q	Charge	C
t	Time	s

3.1.2 Electric Voltage

$$V = \frac{\mathrm{d}W}{\mathrm{d}Q}.$$

Symbols

V	Voltage	V
W	Work	J
Q	Total charge	C

3.1.3 Electric Resistance

$$R = \frac{V}{I}.$$

Symbols

R	Radius	m
V	Voltage	V
I	Current	A

3.1.4 Current Density

$$J = \frac{I}{A}.$$

Symbols

J	Current density	$\mathrm{A/m^2}$
I	Current	A
A	Cross-sectional area	$\mathrm{m^2}$

3.1.5 Resistivity

$$\rho = \frac{E}{J}.$$

Symbols

ρ	Resistivity	$\Omega\,\mathrm{m}$
E	Electric field strength	$\frac{\mathrm{N}}{\mathrm{C}}$
J	Current density	$\mathrm{A/m^2}$

3.1.6 Conductivity

$$\sigma = \frac{1}{\rho}.$$

Symbols

σ	Conductivity	$\Omega^{-1}\,\mathrm{m}^{-1}$
ρ	Resistivity	$\Omega\,\mathrm{m}$

3.1.7 Resistance From Resistivity

$$R = \frac{\rho L}{A}.$$

Symbols

R	Radius	m
ρ	Resistivity	$\Omega\,\mathrm{m}$
L	Length	m
A	Cross-sectional area	$\mathrm{m^2}$

3.1.8 Ohm's Law

$$I \propto V.$$

Symbols

I	Current	A
V	Voltage	V

3.1.9 Electric Power

$$P = VI = I^2 R = \frac{V^2}{R}.$$

Symbols

P	Electrical power	W
V	Voltage	V
I	Current	A
R	Radius	m

3.2 Kirchoff's Circuit Laws

3.2.1 Kirchoff's Current Law (KCL)

$$\sum_{n=1}^{N} I_n = 0.$$

Symbols

I_n	Current at node n	A
N	Amount of nodes	—

3.2.2 Kirchoff's Voltage Law (KVL)

$$\sum_{n=1}^{N} V_n = 0.$$

Symbols

V_n	Voltage difference between two points	V
N	Amount of points	N

3.2.3 Resistors in Series

$$R_{\mathrm{EQ}} = \sum_{n=1}^{N} R_n.$$

Symbols

R_{eq}	Equivalent resistance	Ω
R_n	Resistance of resistor n	Ω
N	Amount of resistors in series	—

3.2.4 Voltage Divider Rule

$$V_n = \frac{R_n}{\sum_{n=1}^{N} R_n} V_s.$$

Symbols

V_n	Voltage difference over resistor n	V
R_n	Resistance over resistor n	Ω
V_s	Source voltage	V

3.2.5 Resistors in Parallel

$$R_{\text{EQ}} = \frac{1}{\sum_{n=1}^{N} \frac{1}{R_n}}.$$

Symbols

R_{eq}	Equivalent resistance	Ω
R_n	Resistance of resistor n	Ω
N	Amount of resistors in parallel	—

3.2.6 Current Divider Rule

$$I_n = \frac{\frac{1}{R_n}}{\sum_{n=1}^{N} \frac{1}{R_n}} I_s.$$

Symbols

I_n	Current difference over resistor n	A
R_n	Resistance over resistor n	Ω
I_s	Source current	A